Dynamic LEO Satellite Routing Approach Based on Deep Graph Attention and Incremental Evolutionary Reinforcement Learning

Zheheng Rao, *Member, IEEE*, Zhenyu Zhu, Dusit Niyato, *Fellow, IEEE*, Ye Yao, *Member, IEEE*, Yanyan Xu, and Yanyu Cheng, *Senior Member, IEEE*

***Abstract*—Low Earth orbit (LEO) satellite networks are an important component of future 6G. However, due to the unique characteristics of the space environment—such as the complexity in modeling network states and the rapid dynamics of the network topology—existing routing methods often struggle to make appropriate routing decisions in the LEO satellite network context, which significantly limits network transmission performance. In this article, we propose a dynamic satellite routing method based on deep graph attention and incremental evolution strategy (DGA-IES). First, to address the challenge of accurately perceiving satellite network information, we introduce a topological perception learning model based on deep graph attention. By combining an enhanced message passing process with a self-attention mechanism, this model effectively captures complex features of the LEO network state, including intersatellite connectivity relationships, as well as the resource states of satellites and links. Second, to tackle the problem of inefficient routing reconvergence in rapidly changing topologies, this article integrates evolution strategies (ESs) into deep reinforcement learning (DRL) approaches. We use the global parallel processing capabilities of ES to replace the sequential interactive proximal policy optimization (PPO) strategy in existing DRL. Moreover, we design an incremental evolutionary process based on satellite motion patterns, facilitating efficient routing convergence in highly dynamic satellite environments. Experimental results demonstrate that our DGA-IES approach enhances LEO network load balancing by reducing end-to-end (E2E) network latency by 10.3%∼58.1%, decreasing packet loss by 3.8%∼20.0%, and improving throughput by 11.1%∼57.0% compared with the benchmark approaches.**

***Index Terms*—Deep graph attention (DGA), deep reinforcement learning (DRL), incremental evolution strategy (IES), low Earth orbit (LEO) satellite network, routing.**

Received 19 March 2025; revised 13 July 2025 and 11 August 2025; accepted 3 September 2025. Date of publication 8 September 2025; date of current version 20 November 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3902804; in part by the Open Research Fund Program of LIESMARS under Grant 24I05; in part by the National Natural Science Foundation of China under Grant 42271431 and Grant 42271425; in part by the Zhejiang Provincial Natural Science Foundation of China under Award LY24F020017; in part by the Singapore Ministry of Education (MOE) Tier 1 under Grant RG87/22 and Grant RG24/24; in part by the NTU Centre for Computational Technologies in Finance (NTU-CCTF); and in part by the RIE2025 Industry Alignment Fund–Industry Collaboration Projects (IAF-ICP) under Grant I2301E0026, administered by A*STAR. This article was presented in part at the IEEE International Conference on Communications (ICC), Montreal, Canada, Jun. 2025. *(Corresponding author: Yanyu Cheng.)*

Zheheng Rao, Zhenyu Zhu, Ye Yao, and Yanyu Cheng are with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: raozheheng@hdu.edu.cn; zhuzhenyu@hdu.edu.cn; yaoye@hdu.edu.cn; yycheng@hdu.edu.cn).

Dusit Niyato is with the College of Computing and Data Science, Nanyang Technological University, Singapore 639798 (e-mail: dniyato@ntu.edu.sg).

Yanyan Xu is with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan 430072, China (e-mail: xuyy@whu.edu.cn).

Digital Object Identifier 10.1109/JIOT.2025.3607492

I. Introduction

RECENTLY, satellite communication has become a key technology for next-generation wireless communication, playing a crucial role in national security and socio-economic development applications, such as emergency response, navigational positioning, long-distance navigation, and aerospace surveying [1]. This technology provides extensive coverage, long-range communication, and geographical independence [2], [3]. Based on orbital altitudes, satellite networks are classified into low Earth orbit (LEO), medium Earth orbit (MEO), and geostationary orbit (GEO). Among these, LEO satellite constellations have garnered increasing attention from researchers due to their cost-effectiveness, reduced transmission latency, superior coverage, and enhanced communication capacity, positioning them as focal points in satellite network studies. Prominent LEO satellite networks include Starlink, OneWeb, and the Iridium constellation.

Routing is a fundamental function in communication networks that optimizes network resource utilization and enhances performance by efficiently selecting paths for data packet transmission [4]. However, research and deployment efforts have shifted from conventional GEO to LEO satellite networks, where numerous rapidly moving nodes operate in a complex space environment. This dynamic space communication setting presents substantial challenges for existing routing methods, complicating the task of making effective routing decisions that maintain network Quality of Service (QoS) in such conditions.

Conventional routing protocols, such as open shortest path first (OSPF) [5] and ad-hoc on-demand distance vector (AODV) [6], select the shortest path based solely on current network topology, disregarding real-time network state conditions, which limit their ability to make appropriate routing decisions. To address this issue, intelligent routing approaches using deep learning (DL) and reinforcement learning (RL) have been introduced. However, these DL-based methods depend heavily on training data, making it difficult to model unknown network states in dynamic wireless network environments.

Deep RL (DRL), which combines the pattern recognition of DL with the decision-making strengths of RL, has demonstrated considerable potential in optimizing wireless networks. Unlike methods that use only DL or RL, DRL-based intelligent routing methods leverage neural networks for feature extraction and RL for routing decisions, leading to enhanced network status awareness and improved routing decisions. However, most existing DRL-based approaches struggle to adapt to the dynamically changing topologies of satellite networks. These challenges stem from the reliance on neural networks designed for processing Euclidean spatial data, including deep neural networks (DNNs) and convolutional neural networks (CNNs). While these networks excel in handling structured data such as images, they perform poorly with non-Euclidean data inherent in network topologies.

Graph neural networks (GNNs) have recently demonstrated strong capabilities in processing and analyzing graph-structured data. Message-passing neural networks (MPNNs), a key GNN model, use aggregation and update processes to understand the interconnections among graph elements and execute relational reasoning and combinatorial generalization. In network topology, each edge possesses unique features, and MPNNs can aggregate these non-Euclidean neighborhood features to enhance routing decisions [7]. However, due to the over-smoothing problem [8], existing approaches can effectively sense only a limited number of neighbors in large-scale multiorder neighbors correlation LEO satellite routing tasks, restricting accurate network state sensing and limiting routing performance.

In highly dynamic network environments, efficient reconvergence of training routing methods is crucial for maintaining network performance, as delays in policy generation impede load scheduling in rapidly changing conditions. DRL training relies on agent-environment interactions and updating neural network weights through backpropagation. While high-performance hardware such as GPUs can accelerate neural network training through backpropagation, the sequential nature of agent-environment interactions and slow weight updates limit the convergence speed, making it challenging to deploy these effectively in satellite networks.

In summary, existing LEO routing models and methods, such as GNN and DRL, face challenges, such as over-smoothing and poor convergence efficiency of modeling and routing in complete LEO satellite networks. To address these challenges, this study proposes a novel approach called the dynamic satellite routing method based on deep graph attention and incremental evolutionary RL (DGA-IES). By utilizing a deep graph attention (DGA) strategy, our model effectively captures network state information, while a parallel incremental evolution strategy (IES) ensures fast convergence speeds of the routing strategy in highly dynamic interplanetary scenarios. The logical framework of this study is shown in Fig. 1. The key contributions of this article are as follows.

1) To tackle the routing optimization objective in highly dynamic and complex intersatellite environments, we propose a multilayer software defined networking (SDN) management architecture comprising GEO, MEO, and LEO satellites. This architecture integrates the advantages of distributed reasoning and centralized decision-making, ensuring the efficiency of model inference within the LEO constellation environment while effectively enhancing routing performance.
2) To address the inaccurate state modeling and over-smoothing problems commonly seen in GNN-based representations of dynamic satellite networks, we propose a DGA model. This model enhances the extraction of non-Euclidean structural features in large-scale LEO constellations and improves state distinguishability through multistep message propagation and attention mechanisms.
3) To overcome the low training efficiency and poor adaptability of traditional DRL methods in dynamic environments, we introduce an IES for DRL training. This approach replaces sequential policy updates with globally parallel evolutionary optimization, significantly accelerating convergence. Moreover, the incremental design enables the model to retain knowledge of periodic structures while adapting rapidly to unpredictable topological changes, thereby supporting fast and robust decision-making in highly dynamic satellite scenarios.
4) We evaluate and compare the DGA-IES with the benchmark solutions by conducting extensive experiments on real LEO network topologies. The experimental results demonstrate that our approach achieves enhanced network load balancing by reducing end-to-end (E2E) network delay by 10.3%~58.1%, lowering packet loss rates by 3.8%~20.0%, and improving throughput by 11.1%~57.0% compared to the benchmark approaches.

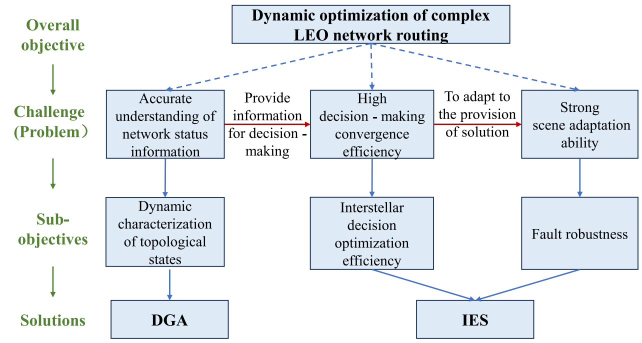


Fig. 1. Logical framework of this study.

The remainder of this manuscript is organized as follows. Section II reviews existing research on LEO satellite routing. Section III describes the system model and routing problem. Section IV outlines the architecture of the DGA-IES methodology and the approach to modeling the state of the interstellar network. Section V elaborates on the principles of intelligent decision generation within dynamic interstellar networks. Section VI details the experimental simulation setup and results analysis. Section VII highlights key challenges and future research directions in intelligent routing for next-generation mobile communication. Finally, Section VIII summarizes this article's contributions.

II. Related Work

This section reviews recent developments in routing technologies for LEO satellite networks and relevant literature to provide context on the challenges and opportunities in this domain.

A. Traditional Routing Protocols

Satellite networks are characterized by high-dynamic and unstable conditions, distinguishing them from terrestrial networks. Consequently, traditional routing methods are not suitable for satellite network environments. These traditional routing methods typically assume static or quasi-static network topologies, where the frequency of topology changes is relatively low. However, in LEO satellite networks, the high-speed movement of satellites, frequent link changes and interruptions result in constantly shifting network topologies. Traditional routing methods face increasing difficulty in adapting to such rapidly changing environments and making timely optimal routing decisions. For example, OSPF [5] require periodic updates of the entire network topology. However, due to frequent changes in link states, this method can lead to delays and inconsistencies. Similarly, AODV [6] may suffer from long route discovery times and higher overhead when network topologies are unstable.

To address these challenges, there is an urgent need to develop routing methods that can dynamically adapt to the unique characteristics of LEO satellite networks. Such methods should be capable of making real-time routing decisions based on satellite network state and link availability. Additionally, to cope with the constantly changing satellite topology, these methods must also accelerate the convergence of routing algorithms to provide more reliable communication in such highly mobile environments.

B. Reinforcement Learning-Based Routing Protocols

The application of RL to the LEO satellite routing problem focuses on modeling the transmission process as a Markov decision process (MDP), where states, actions, and rewards are defined based on the network scenario. Agents learn from interactions with the environment to make routing decisions suited to the current network state [9], [10], [11]. Several RL-based LEO satellite routing methods have been developed in recent years. In [12], Wang et al. employed the Q-routing policy to train agents iteratively, enabling them to identify optimal paths to the destination node and establish the initial routing function. In [13], Mahboob devised a distributed Q-learning-based multiagent RL method to enhance routing efficiency in dynamic environments by deploying agents at each satellite node. In [14], Soret et al. modeled the LEO satellite routing problem as a multiagent partially observable Markov decision problem (POMDP) and developed a distributed Q-learning approach that leverages neighboring node information to enhance decision-making at each node. In [15], Lyu et al. formulated the routing problem as a max–min optimization and proposed a multiagent RL algorithm, which updates distributed routing policies and Lagrange multipliers to balance E2E delay reduction while complying with the constraints.

The RL-based satellite routing algorithms mentioned above explore action-value functions for routing decisions through agent training. However, these algorithms typically store and update these functions in a tabular format, leading to excessive storage and execution overhead in high-dimensional state spaces common in satellite routing. Moreover, traditional RL algorithms lack the capability to model complex network state features, making them unsuitable for the dynamic and large-scale environments of LEO satellite networks.

C. Deep Reinforcement Learning-Based Routing Protocols

Recent advancements in DRL techniques have led some researchers to implement this technique to develop routing schemes for LEO satellites. The DRL approach integrates the cognitive performance of DL with the decision-making capability of RL, allowing for the learning of additional complex strategies to tackle routing optimization challenges involving larger state and action spaces and intricate optimization objectives [16]. In [17], Magadum et al. integrated DRL with SDN to optimize satellite routing based on QoS as a performance parameter. While this approach achieves lower latency and higher throughput, conventional neural networks often fail to capture network features of dynamic, non-Euclidean spatial structures, potentially degrading network performance. In [18], Huang et al. proposed a DRL method that combines the deep Q-network (DQN) algorithm for discrete resource allocation with the proximal policy optimization (PPO) algorithm for continuous power control, aiming to optimize resource sharing and rate-split multiple access. However, the PPO algorithm's reliance on conventional backpropagation hinders its ability to meet the rapid convergence needs of satellite networks. Similarly, Dong et al. [19] introduced a DQN-based load balancing routing technique for LEO satellites. By utilizing a model obtained through offline DQN training, satellite nodes can select optimal routing paths based on delay, bandwidth, and queue utilization of the surrounding satellite nodes. However, due to the dynamic nature of satellite topology, the model's performance may decline following changes in network structure, despite its previous optimal routing decisions.

While intelligent routing schemes for LEO satellite network adaptation have garnered substantial interest from researchers, many of the associated challenges remain unresolved. Conventional neural networks excel in processing Euclidean spatial data, such as images, but struggle with graphical data such as network topology [20], [21], prompting the exploration of applying GNNs for LEO satellite routing. Recent studies [22], [23], [24], [25] have advocated integrating GNNs into LEO satellite networks through routing approaches that combine MPNNs with DRL. Some advanced research has employed more complex GNN models alongside DRL, such as GraphSAGE [26], facilitating the dynamic identification of optimal routing paths for LEO satellite networks. However, traditional MPNNs face the over-smoothing problem [27], which hinders the effective capture and characterization of network features in large-scale constellations. In contrast, current DRL approaches encounter scalability challenges. The inefficient reconvergence strategy employed in these methods results in delays in generating routing strategies, failing to keep pace with the dynamic cycles of LEO constellations. This lag adversely impacts the routing strategy's ability to adapt to evolving network scenarios, significantly impacting network performance.

To address these issues, we propose a routing method that integrates two components: 1) DGA, which enhances message passing in MPNNs to alleviate over-smoothing and model complex non-Euclidean LEO topologies and 2) IES, which adopts a parallel evolutionary strategy for faster policy updates. An incremental evolutionary process further leverages LEO's periodicity while adapting to unpredictable link failures for rapid convergence. Furthermore, the central idea of this work also shows potential applicability to other dynamic network scenarios beyond LEO satellite systems. The graph attention mechanism for handling non-Euclidean data, and the parallel evolutionary strategy in dynamic environments, can be generalized to other networks with dynamic topologies and complex connectivity (such as vehicular networks and mobile ad hoc networks). By adjusting topology-aware parameters and state representations, the method retains its efficient decision logic while adapting to diverse dynamic environments.

III. System Model and Problem Formulation

A. Overall Architecture

The objective of dynamic routing methods for LEO satellite networks is to identify optimal combinations of satellite nodes and intersatellite links to transmit data from the source node to the destination node within a complex and dynamic network environment. The mathematical notations used in this study are detailed in Table I.

TABLE I
Notation Definition

Notation	Description
a_{il}	The attention of node i to the message from step l
$\mathcal{A}$	The action space
$bandwidth$	The traffic demand
B_e	The link bandwidth of edge e
C_e	The link centrality of edge e
dst	The destination node
$\mathcal{D}$	A dynamic satellite network
$e(v_i, v_j)$	The edge between v_i and v_j
$f(x)$	The fitness function
$\underline{f_q}$	The average number of flows queued
F	The Fisher information matrix
$G = (\mathcal{V}, \mathcal{E})$	The network with node set $\mathcal{V}$ and edge set $\mathcal{E}$
G_{ar}	The receive antenna gain
G_{at}	The antenna gain
H_i^t	The node's hidden feature in round t
k_B	Boltzmann constant
L_{ij}	The path loss between v_i and v_j
M	The number of time slots
M_t	The message function
n	The number of agents
N_l	The l-order neighbors
o	The occupancy rate of the satellite node
$p(x\|\theta)$	The search distribution density
P_0	The routing objective function
P_0	The probability of the satellite node being idle
$P_{bandwidth}$	The traffic demand of flow P
P_e	The number of E2E paths traversing the edge e
P_{total}	The total number of paths in the network
Q_v	The buffer queue of satellite node
Q_{max}	The maximum buffer queue allowed for satellite node
R_{ij}	The theoretical maximum transmission rate between v_i and v_j
R_{max}	The maximum data transmission rate
$\mathcal{R}$	The reward function
r_{fa}	The flow arrival rate at the satellite node
src	The source node
$\mathcal{S}$	The state space
T	Iteration rounds required for convergence
T_{ij}^{total}	One-hop delay for v_i and v_j
T_{max}	The limited maximum delay
T^f	The forwarding delay
T_{ij}^p	The propagation delay between v_i and v_j
T_i^Q	The queuing delay of v_i
T_{path}	The total delay for routing
T_n	Thermal noise
$\mathcal{T}$	The operation period of the satellite
τ	A time slot
τ_{length}	The length of a time slot
U_t	The update function
v_i	The i-th LEO satellite node
ω_{max}	The maximum transmission power
ω_{tr}	The transmit power
$\mathbf{W}_a$	The learnable attention parameter matrix
y_p	The population size
λ	The effective flow processing rate
μ	The forwarding rate
ϵ	Gaussian perturbation vector

This article proposes a method that fully leverages the multilayer architecture of current satellite networks. By introducing SDN technology, the proposed approach decouples the network's control and data planes, enabling centralized and programmable control over the entire network. This separation not only facilitates global optimization in routing decisions but also supports flexible, fine-grained traffic management, and dynamic resource allocation [28]. Moreover, the SDN architecture allows for remote software updates and rapid service deployment, significantly improving the network's adaptability to dynamic topology changes and diverse application requirements [29], [30].

As illustrated in Fig. 2, the proposed software-defined satellite network architecture consists of a logically centralized control plane and a distributed data plane. MEO or GEO satellites, equipped with greater computational and storage capabilities, are designated as control plane nodes, responsible for maintaining global network state, computing routing policies, and orchestrating the behavior of the entire system. Meanwhile, LEO satellites, characterized by low propagation delay and wide ground coverage, serve as the data plane, executing the forwarding tasks based on the flow tables provided by the controllers. This hierarchical SDN-enabled architecture effectively supports scalable, resilient, and autonomous routing in large-scale LEO constellations [31].

Control Plane: The control plane consists of GEO and MEO satellites, which possess strong computing and storage capabilities and are responsible for global network control. Specifically, GEO satellites act as top-level controllers, collecting and aggregating global network state and generating global routing policies. MEO satellites serve as regional controllers that receive and refine the strategies from GEO satellites,

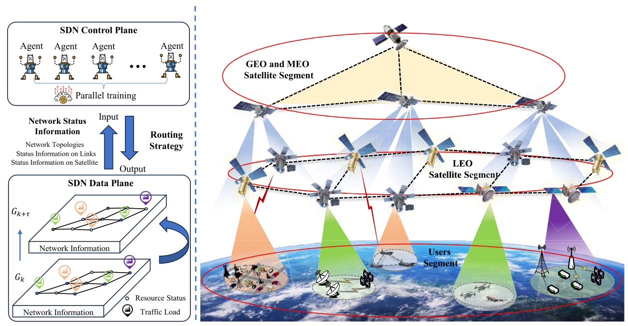


Fig. 2. LEO satellite network model.

coordinate the LEO satellites within their coverage areas, and aggregate local state information reported by LEO nodes.

GEO and MEO satellites collaborate through a hierarchical control mechanism combining top-down policy dissemination and bottom-up state reporting. GEO satellites generate global routing strategies and distribute them to MEO satellites, which refine them into executable flow tables for LEO satellites. In return, LEO satellites periodically—or upon topology changes—report local status to MEO controllers, which aggregate and forward the summarized information to GEO satellites. This design enables the control plane to maintain a global view while remaining responsive to local dynamics.

Data Plane: The data plane consists of LEO satellites and ground stations, and is primarily responsible for data forwarding. LEO satellites execute packet forwarding based on the flow tables issued by the control plane, and continuously report their own status as well as link conditions to the MEO satellites in real time, enabling the control plane to maintain awareness of network dynamics.

The SDN architecture provides crucial support for the DGA-IES model. Real-time state information collected by the control plane forms the input to the DGA module, enabling accurate topology-aware decision-making. For IES, the trained policy can then be efficiently distributed via the GEO–MEO–LEO control hierarchy, ensuring timely and region-specific execution. This centralized training with distributed execution enhances the model's adaptability to dynamic satellite environments.

B. Mobility Satellite Model

Each LEO satellite can establish intersatellite links with up to four neighboring satellites, including two in the same orbit and two in different orbits (Fig. 2). In this study, we denote the set of satellite nodes as $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$, where N represents the total number of satellite nodes. Similarly, the set of intersatellite links is signified as $\mathcal{E} = \{e(v_1, v_2), e(v_2, v_3), \ldots, e(v_i, v_j)\}$, where $e(v_i, v_j)$ denotes the link between satellite nodes v_i and v_j.

The satellite network experiences regular positional changes and occasional link failures or reconstructions, causing its topology to evolve continuously [32]. To facilitate analysis, we implement the principle of virtual topology [33], [34] and divide the operation period of the satellite into equal-length time slots $\mathcal{T} = \{\tau_0, \ldots, \tau, \ldots, \tau_M\}$, where M indicates the total number of time slots. Consequently, we model the dynamic satellite network as a set $\mathcal{D} = \{G_0, \ldots, G_\tau, \ldots, G_M\}$, where, for simplicity, we will denote G_{τ_0} as G_0, and $G_\tau = (\mathcal{V}_\tau, \mathcal{E}_\tau)$ represents an directed graph depicting the satellite network at time τ, fulfilling the following condition:

$$\begin{aligned} G_{\tau+x} &= G_\tau, \quad x \in \left[0, \tau_{\text{length}}\right] \\ G_{\tau+x} &\neq G_\tau, \quad x > \tau_{\text{length}} \end{aligned} \tag{1}$$

where τ_{length} signifies the duration of a time slot. The network topology and link characteristics of the satellite network remain constant throughout each time slot, changing only at the commencement of the subsequent time slot.

C. Channel Model

In the LEO satellite network, data are transmitted through a free-space environment, necessitating consideration of only free-space path loss [35]. The spatial distance between satellites can be determined from their coordinates

$$\left\| e(v_i, v_j) \right\| = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2 + \left(z_i - z_j\right)^2} \tag{2}$$

where $\|e(v_i, v_j)\|$ denotes the spatial distance between v_i, and v_j,(x_i, y_i, z_i) and (x_j, y_j, z_j) represents the spatial coordinates of the satellite nodes v_i and v_j. Subsequently, the path loss between v_i and v_j can be computed as follows:

$$L_{ij} = 20lg\left(\frac{4\pi \left\| e(v_i, v_j) \right\|}{\mathcal{W}}\right) \tag{3}$$

where $\mathcal{W}$ denotes the optical signal wavelength. According to Shannon's theorem, the maximum data transfer rate can be expressed as follows:

$$R_{ij} = B_{ij}\log_2\left(1 + \frac{\omega_{tr} G_{at} G_{ar}}{L_{ij} k_B B_{ij} T_n}\right) \tag{4}$$

where R_{ij} denotes the theoretical maximum transmission rate between nodes v_i and v_j; ω_{tr} signifies the transmit power; G_{at} and G_{ar} represent the transmit and receive antenna gains, respectively; k_B is the Boltzmann constant; B_{ij} indicates the channel bandwidth between the satellite v_i and v_j; T_n refers to the thermal noise.

D. Routing Model

The routing method aims to select the optimal set of nodes and links to transmit each data flow from the source node to the destination node. The flow P is defined as a tuple $(src, dst, bandwidth)$, where src and dst denote the source and destination nodes, respectively; $bandwidth$ represents the required link bandwidth for data transmission. Once a flow is generated, the agent selects the optimal link for transmission based on the current global state and flow requirements.

In the LEO satellite network, the E2E delay is composed of queuing, forwarding, and propagation delays. Each satellite node maintains a buffer queue with a maximum threshold of $Q_{\max}$, operating under the first-in–first-out (FIFO) policy. Therefore, incoming flows at a satellite node must wait for all prior flows to be transmitted. The occupancy rate of the satellite nodes is defined as

$$o = \frac{r_{fa}}{\mu} \tag{5}$$

where r_{fa} denotes the flow arrival rate at the satellite nodes, and μ represents the forwarding rate. Subsequently, the forwarding delay in satellite nodes can be expressed as

$$T^f = \frac{1}{\mu}. \tag{6}$$

The average number of flows queued in the satellite nodes buffer can be expressed as [36]

$$\overline{f_q} = \frac{o}{1-o} - \frac{(Q_{\max}+1)o^{Q_{\max}+1}}{1-o^{Q_{\max}+1}}. \tag{7}$$

Consequently, the effective flow processing rate of satellite nodes can be expressed as

$$\lambda = \mu(1-P_0) \tag{8}$$

where P_0 signifies the probability of the LEO satellite being idle

$$P_0 = \frac{1-o}{1-o^{Q}_{\max}+1}. \tag{9}$$

Therefore, the queuing delay of satellite node v_i can be expressed as

$$T_i^Q = \frac{\overline{f_q}}{\lambda} - \frac{1}{\mu}. \tag{10}$$

When a flow reaches satellite node v_i and is forwarded to the next hop node v_j, the propagation delay between v_i and v_j is determined by the link distance between them

$$T_{ij}^p = \frac{\left\| e(v_i, v_j) \right\|}{c} \tag{11}$$

where c represents the speed of light at which the flow propagate through space.

Based on the (5)–(11), we compute the delay required for one-hop routing for v_i and v_j

$$T_{ij}^{\text{total}} = T^f + T_i^Q + T_{ij}^p. \tag{12}$$

The total delay for completing the routing is

$$T_{\text{path}} = \sum_{e(v_i,v_j)\in \text{mpath}} T_{ij}^{\text{total}} \tag{13}$$

where mpath represents the complete set of routing paths, which includes all intersatellite links required to complete the routing.

E. Problem Formulation

Unlike terrestrial networks, satellite networks exhibit dynamic and complex characteristics, making routing optimization particularly challenging. To ensure continuous and stable operation of the network over an indefinite time horizon, the routing model in practical scenarios must meet two critical objectives: 1) at any given moment, the proposed method should swiftly identify the optimal routing solution, and 2) under any dynamic environmental changes, the proposed approach should rapidly adjust the routing solution and converge to the new optimal state. Therefore, the objective of our routing optimization problem is to ensure high-quality routing decisions within each time slot, while maintaining rapid adaptability to dynamic environments. In this way, the model can achieve stable performance over time through per-slot optimization and continuous adjustment

$$\begin{array}{lll} & \mathcal{P}_0\colon\ \max \text{Traffic}(\tau) & \forall \tau \in (0,\ M] \\ \text{s.t.} & C_1 : \text{convergence time} \le \tau_{\text{length}} & \\ & C_2 : 0 \le R_{ij} \le R_{\max} & \forall e(v_i, v_j) \in \mathcal{E} \\ & C_3 : 0 \le \omega_v \le \omega_{\max} & \forall v \in \mathcal{V} \\ & C_4 : 0 \le Q_v \le Q_{\max} & \forall v \in \mathcal{V} \\ & C_5 : P_{\text{bandwidth}} > 0 & \forall P \\ & C_6\colon T_{\text{path}} < T_{\max} & \forall P \end{array} \tag{14}$$

where $\mathcal{P}_0$ aims to maximize the total network traffic within a single time slot by determining the optimal routing paths for data transmissions in that slot. Constraint $C1$ ensures the routing model's convergence time is minimized, enabling quick reconvergence within the time slot. Constraints C_2 and C_3 restrict each satellite v's transmission power ω_v and data transmission rate R_{ij} to their respective maximum values ($\omega_{\max}$ and $R_{\max}$), ensure reasonable allocation of satellite power and bandwidth resources to avoid queue overflow caused by node overloading. Constraint C_4 limits the node queue lengths Q of each node v to their maximum values at any time t. Constraint C_5 guarantees that the traffic demand for each flow exceeds zero, and Constraint C_6 ensures routing delays remain within an allowable maximum threshold. These constraints collectively facilitate rapid routing adjustments and convergence, thereby supporting stable and efficient network operation over time.

Due to the complexity of satellite networks, existing DRL approaches fail to effectively extract network characteristics and perform poorly in time-varying satellite topologies. Therefore, this study proposes the DGA-IES approach to tackle these challenges.

IV. Satellite Network Architecture and Considered Strategy for State Modeling

This section outlines the training and running processes of the proposed DGA-IES approach and provides a detailed description of the DGA strategy for modeling complex dynamic intersatellite networks.

A. Multilayer Satellite Network Architecture

The proposed DGA-IES intelligent routing algorithm operates within the control plane and is responsible for determining a forwarding path for each flow in the LEO satellite network based on a global view of the data plane through information interaction.

The DGA-IES approach addresses the routing problem in two phases: 1) model training and 2) model running (Fig. 3). The model training phase involves an offline parameter adaptation of the routing model, while the routing decision phase focuses on generating suitable routing decisions based on varying network states.

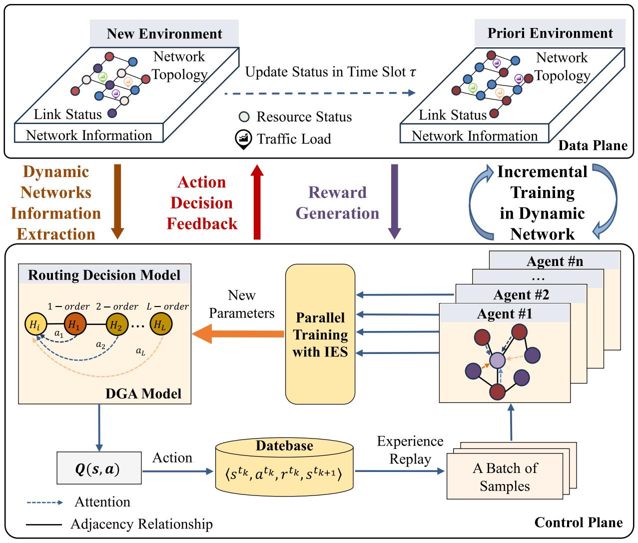


Fig. 3. Diagram of the considered DGA-IES model.

Model Training Phase: Typically, satellites have limited computing resources. Therefore, the training phase is usually carried out offline with the assistance of the powerful computing capabilities of the ground stations. Specifically, the SDN control plane agent processes routing requests at each time step, utilizing global network topology and link information provided by the network monitor as inputs. The agent models the network state s_t and estimates the Q-value for each potential routing action a_t, receiving the reward function r_t from environmental feedback. After the training is completed, the optimized model can be uploaded to the MEO or GEO satellites, where the agent makes real-time decisions based on the current network status.

State Space $\mathcal{S}$: The network state encompasses the current satellite topology and specific link attributes, such as link bandwidth and link centrality

$$\begin{aligned} &\mathcal{S} = \{B_0, \ldots, B_e; C_0, \ldots, C_e\} && \forall e \in E \\ &C_e = \frac{P_e}{P_{\text{total}}} && \forall e \in E \end{aligned} \tag{15}$$

where B_e represents the available bandwidth on link e, and C_e denotes the number of potential paths passing through the edge e, computed as the ratio of E2E paths traversing edge P_e to the total number of paths in the network P_{total}.

Action Space $\mathcal{A}$: The action involves selecting an E2E path connecting source and destination nodes from k candidate paths (e.g., the k shortest paths). Although the state space is defined at the link level, each candidate path is represented by aggregating the state features of its constituent links. This allows the agent to evaluate E2E path quality based on link-level observations

$$\mathcal{A} = \{\text{path}\}_k. \tag{16}$$

Reward Function $\mathcal{R}$: The reward is determined by the agent's decision effectiveness

$$\mathcal{R} = \begin{cases} \text{bandwidth}, & \text{if satisfy the traffic demand} \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

where *bandwidth* is defined as the bandwidth required for the flow. When the agent successfully selects a routing path that meets demands, it receives a reward equivalent to the current traffic demand's bandwidth. In contrast, if the chosen path violates constraints of (14), such as node buffer overflow or exceeding maximum latency, the agent will receive zero rewards.

Model Running Phase: In this phase, routing paths are generated using the optimal weight model obtained during training.

Accurately determining state s_t and evaluating the Q-value in action space a_t are essential for enhancing routing performance. To address the complexities of the LEO satellite network, the DGA is designed to efficiently simulate and analyze dynamic network data within LEO satellite constellations, while the IES enables efficient and intelligent routing decisions.

B. Interstellar Network Modeling Based on Deep Graph Attention Network

The proposed DGA model is designed to accurately capture the structural and dynamic characteristics of large-scale, complex LEO networks while addressing the limitations of traditional GNNs, particularly the over-smoothing problem. For non-Euclidean structured data, GNNs are the leading framework, typically outperforming other approaches. As a prominent GNN model, MPNN improves routing decisions by aggregating features from multiorder neighbor edges, providing comprehensive information. MPNN operates in two phases: message passing and readout. During message passing, each node's hidden feature H_i^t is updated based on messages from neighboring nodes in round t, iterating through T rounds and relying on the message function M_t and update function U_t

$$\overline{m}_i^{t+1} = \sum_{j \in N_1(i)} M_{t+1}\left(H_i^t, H_j^t\right) \tag{18}$$

where each node i receives a message from its immediate neighbor, denoted as message $\overline{m}_i^{t+1}$. Here, neighbors are defined as adjacent links that share a common endpoint with link i; H_i^t and H_j^t represent the hidden features of nodes i and j, respectively, after T iterations; H_i^0 signifies the initial state of node i; $j \in N_1(i)$ refers to the one-hop neighbor of i; The function $M_{t+1}(\cdot, \cdot)$ defines the message exchanged between nodes i and j during round $t+1$ based on their features.

Conventional MPNNs [37] struggle in satellite network routing due to their limitations in accurately modeling large networks, required by LEO satellite network routing tasks. As the number of MPNN layers increases, node features become excessively similar, leading to feature convergence and difficulty distinguishing between different nodes, a problem known as over-smoothing. While deeper networks increase the range of messaging, they often cause all nodes to receive the same information, exacerbating over-smoothing.

Therefore, this study proposes a DGA network that improves satellite network modeling, enhances message passing, and prevents over-smoothing by introducing two key modifications: 1) multistep message sources and 2) multistep attention.

Multistep Message Source: In a single aggregation process, we enlarge the sources of incoming information and connect them by means of a residual structure. As a result, the scope of information interaction is no longer confined to the neighborhood of a node, but extends to the aggregation of multilevel information for a given step size l. Specifically, we expand the messages field beyond a node's immediate neighbors, allowing interactions up to a defined step size l. This step size ranges from 0 to L (L is the maximum step size without over-smoothing), where $l = 0$ indicates the node only communicates with itself. The enhanced message function is expressed as follows:

$$\overline{m}_i^{t+1} = \sum_{l=0}^{L}\left(\sum_{j\in N_l(i)} M_{t+1}\left(H_i^t, H_j^t\right)\right) \tag{19}$$

where each node i receives a message from its l-order neighbor node, $j \in N_l(i)$ denotes the l-order neighbor of i.

Multistep Attention: In satellite networks, nodes have distinct roles in diverse tasks, making it pivotal to adaptively differentiate these roles for efficient resource allocation and improved network modeling. To address this challenge, we introduce a self-attention mechanism to control the aggregation of node-specific messages across various step lengths. We further extend the message-passing paradigm by enabling more refined differentiation as follows:

$$m_i^{t+1} = \sum_{l=0}^{L} a_{il}\left(\sum_{j\in N_l(i)} M_{t+1}\left(H_i^t, H_j^t\right)\right) \tag{20}$$

where a_{il} denotes the attention of node i to the message from step l. It is generated by the learnable parameter matrix $\mathbf{W}_a \in \mathbb{R}^{N\times L}$ and normalized by softmax

$$a_{il} = \frac{e^{\mathbf{W}_a[i,l]}}{\sum_{l'=0}^{L} e^{\mathbf{W}_a[i,l']}} \tag{21}$$

where $a_{il} > 0$ and $\sum_{l=0}^{L} a_{il} = 1$. This mechanism allows each node i to dynamically adjust its attention to the neighborhood messages of different steps.

Although these two designs may introduce additional synchronization and computational overhead, it significantly improves state awareness and routing accuracy in dynamic topologies. We assume that the multistep aggregation can be completed synchronously within each time slot, as the overhead can be substantially reduced through parallel processing and high-performance computing. This assumption is aligned with practical capabilities—modern satellite control infrastructures, equipped with distributed computing resources, can perform the required graph operations within the typical topology update intervals of LEO networks.

Each node i combines its aggregated message m_i^{t+1} with its existing hidden features H_i^t, and the update function computes new hidden features

$$H_i^{t+1} = U_{t+1}\left(H_i^t, m_i^{t+1}\right) \tag{22}$$

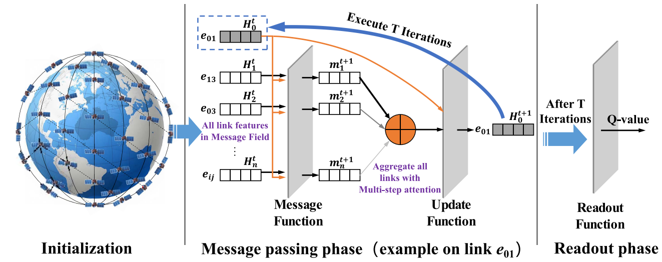


Fig. 4. Message pass architecture in DGA.

Algorithm 1 Improved Message Passing

Input: LEO network topology $G = (\mathcal{V}, \mathcal{E})$, Link state set H.
Output: q-value

1: **for** each $i \in \mathcal{E}$ **do**
2: Initialize H_i^0
3: **end for**
4: **for** $t = 1$ to T **do**
5: **for** each $i \in \mathcal{E}$ **do**
6: $m_i^{t+1} = \sum_{l=0}^{L} a_{il}\left(\sum_{j\in N_l(i)} M_{t+1}\left(H_i^t, H_j^t\right)\right)$
7: $H_i^{t+1} = U_{t+1}\left(H_i^t, m_i^{t+1}\right)$
8: **end for**
9: **end for**
10: $r \leftarrow \sum_{i\in\mathcal{E}} H_i$
11: $q \leftarrow R(r)$

where $U_{t+1}(\cdot,\cdot)$ defines an update function for the hidden features of node i in the t+1 round based on its hidden features and message from neighboring nodes.

Upon completing the modeling process and conducting T rounds of message passing and state updates, each link node gains a comprehensive understanding of the satellite network's overall information state through the aggregation of hidden features from neighbors in the T layer. This aggregated information is then utilized to compute feature vectors for the entire graph, which aims in subsequent decision-making via the readout function in the readout phase. The readout process can be expressed as follows:

$$q = R\left(H_i^T | i \in \mathcal{E}\right) \tag{23}$$

where $\mathcal{E}$ denotes the set of links of the satellite network. The resulting value of q reflects the current action derived from DGA predictions, which will be used as input for IES routing training and decision making.

The message passing architecture of the DGA is illustrated in Fig. 4. In routing tasks, the primary determinant of path selection is the variability in satellite network link properties. Therefore, this study leverages link entities as nodes in the GNN and facilitates message passing among all links. Algorithm 1 outlines the detailed algorithmic flow. The algorithm takes the LEO network topology and the set of link states as the input to generate a q-value for action prediction. Lines 1 and 2 iterate over all links in the satellite network to establish the initial hidden features based on each link's current state. The message-passing process occurs from lines 3 to 6 for a total of T iterations. In each iteration, the message function M_t

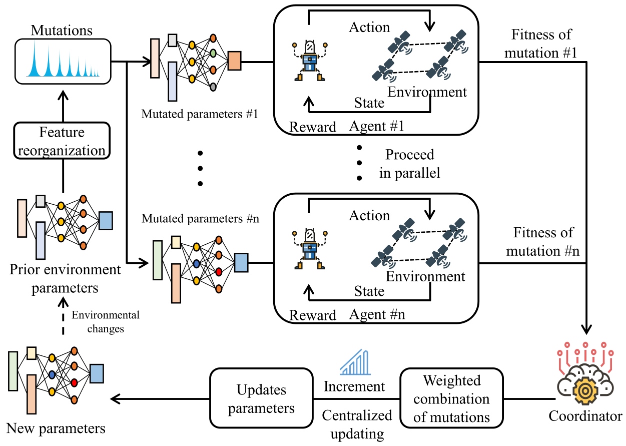


Fig. 5. Incremental evolutionary DRL strategy.

operates on the hidden features H of node i and its neighbors within L steps to produce a new feature vector m_i^{t+1}. The update function U_t is subsequently employed to integrate this new feature vector with the node's existing hidden features. Both the message function M_t and the update function U_t are trainable components. The output from the update function U_t is aggregated in line 7. Finally, the readout function is applied in line 11 to generate the q-value for the current action on the current state. The complexity analysis of Algorithm 1 is discussed in Appendix A. In the worst case scenario, the time complexity of Algorithm 1 is $O(T|\mathcal{E}|)$, which satisfies the requirement for rapid intersatellite convergence. Moreover, the multistep message source and multistep attention mechanisms enhance DGA's performance. Therefore, the designed DGA can achieve competitive performance in an acceptable running time.

V. Dynamic Decision Making Based on Parallel Incremental Evolution Strategy

This section presents the other aspect of the proposed DGA-IES approach—efficient and adaptive routing decisions enabled by the IES, which improve training efficiency and enhance adaptability to dynamic satellite network environments.

Current DRL routing decision methods depend on conventional backpropagation, where the DRL agent learns through sequential interactions with the environment. As task complexity increases, computational and time overheads grow significantly [38], making these methods less suitable for complex and dynamic LEO satellite networks. The evolution strategies (ESs) achieves highly parallel optimization of neural network weight sets by emulating biological evolution principles. This approach alleviates time-consuming computations, such as time discounting, gradient backpropagation, and value function approximation, facilitating optimal solutions [39].

Fig. 5 depicts the IES designed for DRL within the LEO satellite network. The IES enhances routing by applying ES to perturb (mutate) and cross-restructure initial neural network strategy parameters, generating parameters. Individuals with better performance are selected to form the next generation, thereby improving the agents' reward payoffs. The core concept involves iteratively updating search distribution parameters using the sampling gradient of expected fitness.

Let θ denote the parameter of the search distribution density $p(x|\theta)$, and $f(x)$ represent the fitness function (the received reward). The expected fitness in each generation under the search distribution is expressed as follows:

$$J(\theta) = \mathbb{E}_\theta\big[f(x)\big] = \int f(x)p(x|\theta)dx. \tag{24}$$

Similar to the reinforce method [40], the gradient steps for θ are estimated as follows:

$$\begin{aligned}\nabla_\theta J(\theta) &= \nabla_\theta \int f(x)p(x|\theta)dx \\ &= \mathbb{E}_\theta\big[f(x)\nabla_\theta \log p(x|\theta)\big].\end{aligned} \tag{25}$$

The search gradient estimate of the population $(x_1, x_2, \ldots, x_{y_p})$ is obtained using the Monte Carlo strategy

$$\nabla_\theta J(\theta) \approx \frac{1}{y_p}\sum_{i=1}^{y_p} f(x_i)\nabla_\theta \log p(x_i|\theta) \tag{26}$$

where y_p signifies the population size, and the search gradient is estimated in each generation to improve expected fitness.

This study discovers that in complex satellite networks, characterized by ridges and plateaus, using the natural gradient instead of stochastic gradient updates improves model convergence. The natural gradient is guided by the Fisher information matrix associated with specific parameters of the search distribution

$$F = \int p(x|\theta)\nabla_\theta \log p(x|\theta)\nabla_\theta \log p(x|\theta)^T dx \tag{27}$$

where the Fisher information matrix F reflects the sensitivity of θ to the probability distribution $p(x|\theta)$. This matrix quantifies the uncertainty of parameter θ by assessing how sensitive it is to changes in the probability distribution. This matrix captures uncertainty in various directions. For convenience, F can be approximated through mathematical expectations

$$F = \mathbb{E}_\theta\big[\nabla_\theta \log p(x|\theta)\nabla_\theta \log p(x|\theta)^T\big]. \tag{28}$$

If F is invertible, the gradient is adjusted using the inverse of the Fisher information matrix, the natural gradient amounts to

$$\widetilde{\nabla}_\theta J(\theta) = F^{-1}\nabla_\theta J(\theta). \tag{29}$$

To efficiently explore complex high-dimensional satellite network scenarios, the population $\{x_i\}_{i=1}^{y_p}$ is initialized as a multivariate Gaussian centered on θ, with a diagonal covariance matrix. Specifically, $x_i = \theta + \sigma\epsilon_i$, where σ denotes the noise standard deviation and ϵ_i signifies a random variable drawn from a standard Gaussian distribution [41]. The gradient estimator is then expressed as

$$\nabla_\theta \mathbb{E}_{\epsilon\sim\mathcal{N}(0,I)}\big[f(\theta+\sigma\epsilon)\big] = \frac{1}{\sigma}\mathbb{E}_{\epsilon\sim\mathcal{N}(0,I)}\big[f(\theta+\sigma\epsilon)\epsilon\big] \tag{30}$$

which can be estimated with samples

$$\nabla_\theta f(\theta) \approx \frac{1}{y_p \sigma} \sum_{i=1}^{y_p} f(\theta + \sigma \epsilon_i)\epsilon \tag{31}$$

where θ is updated iteratively until convergence using $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$, with α representing the learning rate. The gradient estimation is simplified through a Gaussian perturbation vector $\epsilon \sim \mathcal{N}(0, I)$ under the sampling unit, allowing for the evaluation of the perturbation strategy's performance in adapting to the environment. The evaluated results are then aggregated into the search instance population.

Synchronizing the random seeds among agents before policy optimization allows each agent to anticipate the perturbations applied to others. At this stage, communication among agents is reduced to transmitting a scalar (fitness) for consistent parameter updates, enabling high parallelizability (Fig. 5).

For constraint C_1, given the dynamic nature of satellite networks, model reconvergence must be optimized to handle the continuously evolving network environment. Therefore, a step-incremental evolution model is proposed for the time-domain update approach within the ES. This model assigns higher weights to instances with more new information during parameter updates, encouraging the search distribution to explore new promising regions in the parameter space. ES uses the optimal policy parameters from the previous iteration as the starting parameters for the current moment. The novelty weight can be expressed as

$$\omega(\mathcal{Z}_i) = y_p \cdot \frac{e^{N(\mathcal{Z}_i)/\rho}}{\sum_{j=1}^{y_p} e^{N(\mathcal{Z}_j)/\rho}}, i = 1, \ldots, y_p \tag{32}$$

where $\{\mathcal{Z}_i\}_{i=1}^{y_p}$ is typically instantiated as a multivariate Gaussian with a diagonal covariance matrix centered at θ, i.e., $\mathcal{Z}_i = \theta + \sigma\epsilon_i$; $N(\mathcal{Z}_i)$ denotes the novelty of the parameter $\mathcal{Z}_i$, which is quantified by its deviation from the previous one; ρ signifies the progress factor; ρ increases after each iteration, gradually diminishing the influence of the novelty weight as evolution proceeds, indicating less exploration in the routing strategy over time. By utilizing incremental learning mechanisms, the IES-based method can rapidly adapt to the dynamic topology of LEO satellite networks and continuously optimize routing strategies based on historical experience, thereby ensuring the sustained and reliable long-term operation of the routing algorithm.

Accordingly, the expected fitness is modified as follows:

$$\begin{aligned} J(\theta) &= \mathbb{E}_\theta\big[w(x)f(x)\big] \\ &= \int w(x)f(x)p(x|\theta)dx. \end{aligned} \tag{33}$$

Consequently, the gradient estimator in (20) becomes

$$\nabla_\theta f(\theta) \approx \frac{1}{y_p \sigma} \sum_{i=1}^{y_p} w(\theta + \sigma\epsilon_i) f(\theta + \sigma\epsilon_i)\epsilon_i. \tag{34}$$

The algorithm approximates the gradient in the parameter space by moving toward instances that yield high fitness $f(\theta + \sigma\epsilon_i)$ and significant weighting $w(\theta + \sigma\epsilon_i)$. When the weighting metric accurately reflects the amount of new knowledge in the instances, the gradient estimator rewards those with more new knowledge, guiding the search distribution toward promising regions in the parameter space. This instance-weighting mechanism enhances the algorithm's evolutionary process of searching for effective strategies in new environments, improving adaptability to dynamic learning environments.

To address the resource constraints of LEO satellites and the need for rapid algorithm convergence, it is imperative that the algorithm should minimize the computational load and time-consuming of model training. Therefore, we designed a centralized-distributed approach for training and updating the ES algorithm. In this approach, each local agent interacts independently with the environment. A centralized agent called coordinator, aggregates local information from other agents, including evaluation scores and mutations, computes new policy parameters, and distributes these updated parameters to all agents. Assume that there are n agents training in parallel, fully connected information would require n^2 interactions. The centralized-distributed approach reduces this number to n interactions, lowering the algorithmic complexity from $O(n^2)$ to $O(n)$ [42]. Consequently, the training process for the routing decision model becomes more efficient and energy-saving.

During training, each agent perceives the environment through state vectors generated by DGA. The more accurate the DGA representations are, the more reliable the gradient estimation becomes, enabling IES to perform more efficient parameter updates. In turn, IES evolves the policy parameters θ, which guide the adaptation of DGA's attention weights, making feature extraction more suited to dynamic topologies. The DGA output q and the IES parameters θ follow the mapping $\theta_{t+1} = g(\theta_t, q_t)$, where $g(\cdot)$ denotes the IES update function, enabling the co-evolution of feature representation and decision policy.

The IES algorithm is outlined in Algorithm 2. In this study, the dynamic environment $\mathcal{D}$ is modeled as an LEO satellite network G_τ at time τ. IES is used to train policy parameters θ, and outputs the neural network weights θ_τ^* with optimal performance at time τ. At $\tau = 0$, the ES evolves from scratch (lines 2–17) and iteratively updates policy parameters until the model converges. In each iteration, agents randomly sample perturbations from a Gaussian distribution to mutate the strategy parameters and interact with the environment to obtain a reward score of this mutation. Each agent sends these scores to the coordinator after each mutation for aggregation. The coordinator computes $\Delta\theta$ and distributes it back to the agents, completing the strategy parameter update for that round.

For moments $\tau \geq 1$, the ES switches to incremental training. Lines 19–37 describe the steps where the ES performs this process. Since previously optimal parameters might have overfitted to the original environment, leading to local optima, this could degrade performance in new evolutionary cycles [43]. To overcome this drawback, this study incorporates novelty exploration during incremental training. When an agent generates a new strategy parameter through mutation, it evaluates the novelty weight ω_i of this mutation and sends it to the coordinator with the reward score r_i. The coordinator then

Algorithm 2 IES

Input: Dynamic satellite environment $\mathcal{D} = \{G_0, \ldots, G_\tau, \ldots, G_M\}$, Current period $\tau (\tau \geq 0)$, Learning rate α, Noise standard deviation σ, Increments of progress factor $\Delta\rho$, Initial policy parameters θ.
Output: The optimal policy weight θ_τ^* for G_τ.

if $\tau = 0$ **then**
  while *not converged* **do**
    for *each agent* $i = 1, \ldots, n$ **do**
      Sample $\epsilon_i \sim \mathcal{N}(0, I)$
      Compute local returns $r_i \leftarrow r(\theta_\tau + \sigma\epsilon_i)$
      if *agent* i *is coordinator* **then**
        Receive and aggregate returns of other agents: $\hat{r}$
        Parameters update $\Delta\theta \leftarrow \alpha \frac{1}{n\sigma} \sum_{j=1}^{n} \hat{r}_j \epsilon_j$
        Send $\Delta\theta$ to all the other agent
        Update parameters $\theta_{\tau+1} \leftarrow \theta_\tau + \Delta\theta$
      else
        Send returns r_i to coordinator
        Receive $\Delta\theta$ from coordinator
        Update parameters $\theta_{\tau+1} \leftarrow \theta_\tau + \Delta\theta$
      end if
    end for
  end while
else
  Obtain $\theta_{\tau-1}^*$
  while *not converged* **do**
    for *each agent* $i = 1, \ldots, n$ **do**
      Sample $\epsilon_i \sim \mathcal{N}(0, I)$
      Compute local returns $r_i \leftarrow r(\theta_\tau + \sigma\epsilon_i)$
      Calculate and normalize weights $\omega_i \leftarrow \omega(\theta_\tau + \sigma\epsilon_i)$ by (32)
      if *agent* i *is coordinator* **then**
        Receive and aggregate returns and weights of other agents: $\hat{r}$, $\hat{\omega}$
        Parameters update $\Delta\theta \leftarrow \alpha \frac{1}{n\sigma} \sum_{j=1}^{n} \hat{\omega}_j \hat{r}_j \epsilon_j$
        Send $\Delta\theta$ to all the other agent
        Update parameters $\theta_{\tau+1} \leftarrow \theta_\tau + \Delta\theta$
      else
        Send returns r_i,weights ω_i to coordinator
        Receive $\Delta\theta$ from coordinator
        Update parameters $\theta_{\tau+1} \leftarrow \theta_\tau + \Delta\theta$
      end if
    end for
    $\rho \leftarrow \rho + \Delta\rho$
  end while
end if

employs both the score r and novelty weight ω to determine $\Delta\theta$, effectively combining novelty with the quality of this mutation for better performance. The complexity analysis of Algorithm 2 is discussed in Appendix B. The parallel training approach of IES provides the model with faster convergence, making it well-suited for dynamically changing satellite network environments. The IES adjusts the search direction by the natural gradient of equation (29) to avoid falling into a local optimum, thus ensuring convergence, the theoretical proof of which is given in the literature [39].

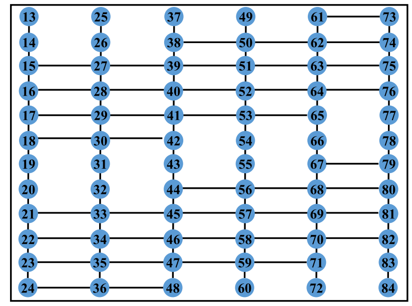


Fig. 6. Iridium topology.

VI. Numerical Results

This section evaluates the performance of the DGA-IES method through numerical analysis across various scenarios, including hyperparameter analysis, performance assessment, and ablation studies.

A. Experimental Settings

The satellite network environment is simulated using network simulator version 3 (NS3) [44] for the data plane, while the control plane agent is deployed in OpenAI Gym [45]. The source code was written in Python, utilizing libraries, such as NumPy for vector computations [46], TensorFlow for DGA implementation [47], NetworkX [48] for network state representation, and MPI4PY for agent communication [49]. All simulations run on an Intel Xeon W-2145 CPU (3.70 GHz, 16 cores), Ubuntu operating system, Nvidia GPU A100, and 128-GB RAM.

The experiment employs the Iridium satellite topology, depicted in Fig. 6. The topology has a total of 66 satellites in six orbits, and four intersatellite links per satellite, with an orbital height of 780 km and an orbital inclination of 86.4°[19]. Bandwidth and delay values for each satellite link are randomly generated. In the experiment, the flow randomly generates flows for source nodes, destination nodes, and traffic demands. The agent are responsible for routing these flows based on the current network state. The assumption is made that all satellite nodes possess the capability to generate, forward, and process incoming flows.

B. Key Parametric Analysis

1) Determine the Number of Parallel Agents: We evaluated the influence of the number of DRL agents on training time. The training experiments are conducted on the Iridium topology with agent counts of 1, 2, 4, 8, 16, 32, and 64. During each experiment, the time per training iteration is measured, and the average reward after convergence is determined. A higher average reward signifies more effective routing decisions, underscoring the impact of agent counts on the training process for satellite routing.

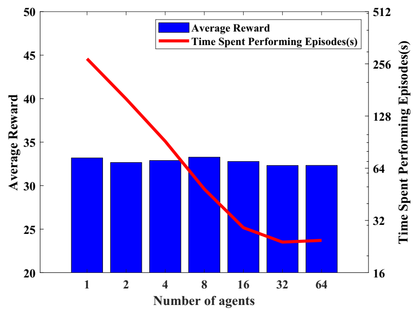


Fig. 7. Relationship between number of agents and training time, average reward.

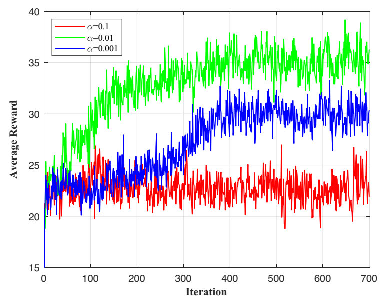


Fig. 8. Relationship between learning rate of agents and average reward (mean reward).

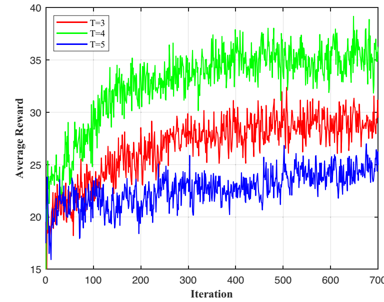


Fig. 9. Relationship between propagation steps and average reward.

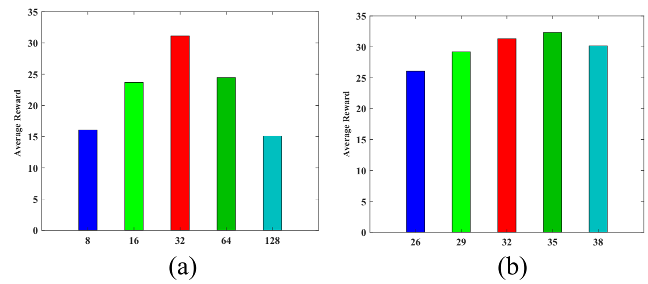


Fig. 10. Relationship between the link state dimensions (readout units) and the mean (average) reward. (a) Initial comparison; (b) Detailed comparison. The horizontal axis represents link state dimensions (readout units), and the vertical axis represents average reward. The determined link state dimension and readout units are mutually correlated and numerically equivalent.

The relationship between agent count, training time, and average reward is depicted in Fig. 7. The horizontal axis represents the agent count processed in parallel, the left vertical axis signifies the bar graph indicating the average rewards achieved after model convergence, and the right vertical axis corresponds to the line graph showing training time per iteration.

The analysis of the graph reveals that training time decreases initially as the agent count increases, reaching a minimum of 32 agents, after which it rises. This pattern occurs because too few agents force each agent to explore a larger area, considerably increasing task complexity and making convergence more difficult. In contrast, too many agents lead to excessive interaction time during agent cooperation, substantially increasing the overall task time overhead. However, the average rewards remain stable across agent counts, indicating that an optimal number of agents enhances the training efficiency of the RL process without compromising training rewards.

Experimental Conclusion: The optimal number of agents improves the efficiency of RL process to explore the optimal solution.

2) Determine Learning Rate of Agents: The reward function reflects the agent's learning progress during DRL training. Fig. 8 illustrates the trend of rewards as the number of iterations increases. The horizontal axis represents the number of iterations explored by the DGA-IES agent, and the vertical axis indicates the average reward obtained. The reward function curves exhibit an initial rise and then stabilize, demonstrating that DGA-IES effectively learns the network environment through the exploration process and converges to optimal routing solutions. The DAR-DRL model yields the highest rewards with a learning rate of 0.01.

Experimental Conclusion: The optimal learning rate effectively prevents the DRL model from converging to local solutions, enabling agents to explore optimal routing solutions.

3) Determine Propagation Steps: The iteration of DGA is determined by the propagation step T. Therefore, this section conducts an experimental analysis to optimize the value of T. Fig. 9 illustrates the impact of different T values on the training performance of the agent. From theoretical perspective, T directly affects the balance between the model's ability to aggregate sufficient information from neighbors and its susceptibility to over-smoothing. If T is too low, the model may fail to capture sufficient network features, leading to underfitting. Conversely, if T is too high, excessive aggregation can cause over-smoothing, where node-specific features are lost, resulting in overfitting. Both scenarios negatively impact the agent's ability to generalize, as evidenced by a decline in the average reward after convergence.

Experimental Conclusion: An appropriate value of T can help the model better fit the network features, thereby improving the average reward.

4) Determine Link State Dimension and Readout Units: To identify appropriate values for the link state dimension and the number of readout units in the policy network, we conduct a two-stage tuning process on the Iridium topology. In the first stage, we evaluate commonly used embedding dimensions from the set {8, 16, 32, 64, 128} and observe the model's final average rewards. Fig. 10(a) indicates that the best performance is achieved when the dimension is around 32, with the optimal range falling between 16 and 64.

Based on this finding, we further narrow the search space in the second stage by testing finer-grained values from the

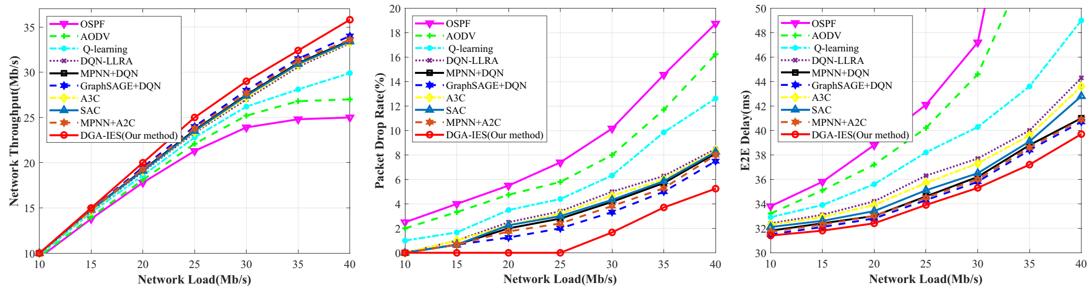


Fig. 11. Performance comparison under different traffic loads.

TABLE II
PARAMETER SETTINGS FOR EXPERIMENT

Parameters	Values
Learning rate α	0.01
Noise standard deviation σ	0.05
Buffer size	5000
Batch size	32
Link State Dimension	35
Readout Units	35
Number of mutations	640
Number of agents n	32
Total training episodes	700
Propagation steps T	4
Maximum step size L	2
Optimizer	Adam

set $\{26, 29, 32, 35, 38\}$. As shown in Fig. 10(b), increasing the dimensionality initially improves performance, as richer embeddings better capture the structure and dynamics of the satellite network. However, beyond a certain point, the performance begins to degrade due to overfitting and reduced convergence speed. Among all tested settings, a dimension of 35 yields the highest average reward and the most stable convergence behavior, effectively balancing representation capacity and training efficiency.

Experimental Conclusion: The routing model achieves optimal performance when the link state dimension and readout unit number are set to 35.

C. Performance Analysis

Nine baseline methods are compared in this study, and all baselines are individually tuned to achieve convergence and optimal performance within the evaluated LEO satellite network environment.

OSPF [5]: The conventional routing method that uses Dijkstra's algorithm to select the shortest routing path.

AODV [6]: The traditional routing method that broadcasts route requests to determine the destination node and shortest path.

Q-Learning [13]: The RL-based algorithm that trains a routing decision model to make routing choices through Q-learning.

DQN-LLRA [19]: The DRL-based algorithm that trains a routing decision model to make routing choices through DQN.

MPNN+DQN [25]: The DRL-based algorithm that uses MPNN to capture relationships between nodes and uses DQN to make adaptive routing decisions in the network.

GraphSAGE+DQN [26]: The DRL-based algorithm that uses GraphSAGE to capture relationships between nodes, and uses DQN to make adaptive routing decisions in the network.

Asynchronous Advantage Actor–Critic (A3C) [50]: The DRL-based algorithm that uses an asynchronous actor–critic architecture to accelerate policy learning for routing decisions.

Soft Actor–Critic (SAC) [51]: The DRL-based algorithm that uses actor–critic architecture with entropy regularization to improve routing decisions.

MPNN+Advantage Actor–Critic (A2C) [52]: The DRL-based algorithm that employs MPNN to extract topological features and uses the A2C framework to make routing decisions in dynamic networks.

1) Dynamic Traffic Scenarios Performance Evaluation: As shown in Fig. 11, the proposed DGA-IES achieves the best overall performance under varying traffic loads. This is due to its ability to accurately model the structural complexity of satellite networks through the DGA module and to rapidly adapt to traffic fluctuations via parallel and incremental training using IES. As a result, DGA-IES significantly improves key performance metrics, such as average throughput, E2E delay, and packet loss rate.

In contrast, traditional methods like OSPF and AODV rely on fixed routing metrics and cannot respond to dynamic traffic changes, leading to poor adaptability and network congestion under fluctuating loads. Q-learning and DQN-LLRA, while more flexible, lack the capacity to model high-dimensional network states, making them less effective in dynamic satellite environments. Actor-critic DRL methods, such as A3C and SAC improve learning stability and exploration, but their sequential training structure limits efficiency and responsiveness in time-sensitive routing tasks. GNN-based approaches like MPNN+DQN, MPNN+A2C, and GraphSAGE+DQN offer stronger structural awareness and perform better, yet they still suffer from over-smoothing and limited high-order feature extraction, which constrain their adaptability.

Experimental Conclusion: By integrating accurate graph-based state modeling and efficient adaptive training, the proposed DGA-IES method outperforms all baselines in dynamic traffic scenarios, demonstrating its strong suitability for real-world satellite network environments.

2) Dynamic Topology Scenarios Performance Evaluation: Based on the previous experimental setup, a representative 25 Mb/s was chosen as the traffic demand for the experiment.

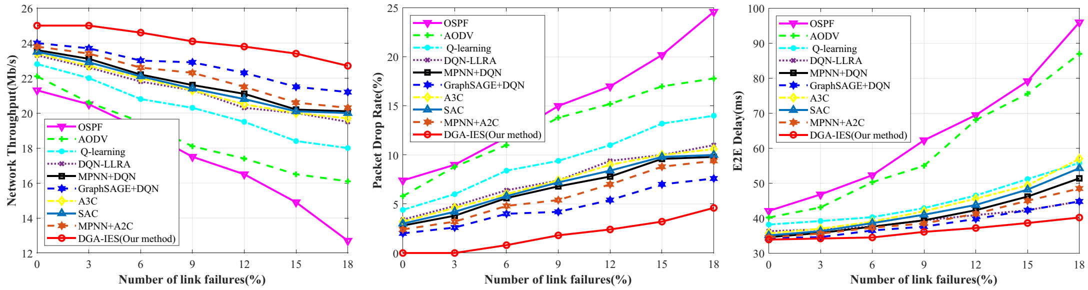


Fig. 12. Performance comparison under link failure scenarios.

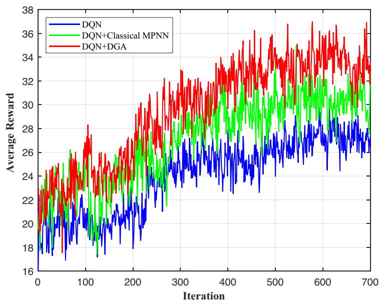


Fig. 13. Convergence curves for DQN, DQN+MPNN, and DQN+DGA.

Afterward, the links on the experimental topology are gradually reduced to simulate a partial link failure scenario.

Results Analysis: The performance comparison link failure scenarios result of DGA-IES with existing methods are presented in Fig. 12. The experimental results show that under the scenario of partial link failures, the proposed DGA-IES has less performance loss in the event of link failures and is more adaptable to dynamically changing network topology compared to other compared methods.

Experimental Conclusion: Compared with other existing methods, the proposed DGA-IES method can make better routing decisions and obtain better network transmission performance in the presence of changing network topology.

D. Ablation Experiments

The ablation experiments aim to evaluate the individual effectiveness of the proposed DGA and IES designs, as well as the performance gain from their integration.

1) Performance of DGA Design: We compared the effectiveness of DRL algorithms using different neural networks, i.e., the DQN method using DNN, the DQN method using Classical MPNN, and the DQN method using DGA.

Results Analysis: Fig. 13 shows the convergence curves of DQN, MPNN+DQN, and DGA+DQN. All three methods exhibit similar learning trends—initial growth followed by convergence—indicating their ability to learn from interactions. However, DGA+DQN achieves significantly higher average rewards. The DQN uses a fully connected neural network, which lacks structural bias and cannot effectively model the non-Euclidean topology of satellite networks. MPNN introduces a graph structure but suffers from oversmoothing, limiting its ability to capture high-order dynamic features. In contrast, DGA employs multistep message source and multistep attention, which enhance representation capability. As shown in Table III, DGA+DQN improves the average reward by 5.6% over MPNN+DQN and 10.5% over DQN.

TABLE III
COMPARISON OF REWARDS FOR DQN, DQN + MPNN, DQN+DGA

Algorithm	DQN + DGA	DQN +MPNN	DQN
Mean Reward	33.22	29.38	26.21

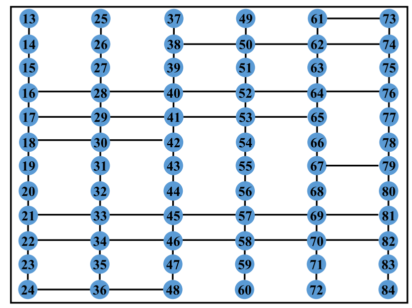


Fig. 14. Iridium-mod topology.

Experimental Conclusion: The DGA module significantly improves topological representation by addressing oversmoothing, providing more informative state inputs for decision-making.

2) Performance of IES Design: Randomly disconnect certain interstellar links of Iridium to simulate random satellite link failures and generate a new experimental topology named Iridium-mod, as illustrated in Fig. 14. Agent test on the Iridium-mod topology after completing the training on the original Iridium topology.

Experimental Scenario 1: Evaluating reconvergence time.

Results Analysis: Fig. 15 compares the convergence time of DQN, ES-DQN, and IES-DQN. The ES-based methods significantly outperform DQN in convergence speed, validating the effectiveness of parallel evolution in accelerating exploration. Among them, ES's parallel training accelerates model training speed, while IES's incremental evolution mechanism reduces the amount of computation required when changing the topology and increases the dynamic adaptability of the algorithm. Specifically, IES-DQN reduces convergence time by 53.87% compared to DQN and by 36.33% compared to ES-DQN.

Experimental Conclusion: IES enhances convergence efficiency by leveraging parallel training to accelerate policy

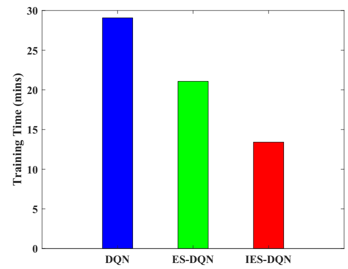
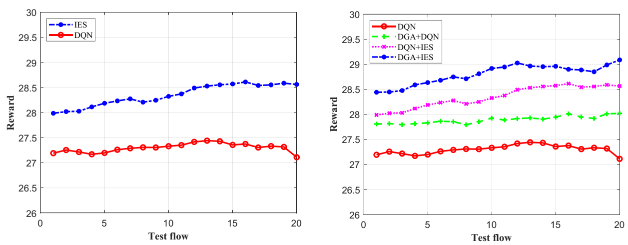


Fig. 15. Comparison of reconvergence times.



(a) (b)

Fig. 16. Performance comparison in a satellite dynamic topology environment: (a) IES and DQN. (b) DGA-IES and baseline algorithms.

evolution, making it well-suited for highly dynamic satellite network environments.

Experimental Scenario 2: Evaluating reconvergence reward.

Results Analysis: Fig. 16(a) shows the average rewards after topology changes. IES-DQN consistently achieves higher rewards than baseline methods, demonstrating its superior adaptability. This improvement stems from the evolutionary mechanism's ability to generate diverse policy variations through mutation and recombination, increasing the likelihood of finding near-optimal solutions. Quantitatively, IES-DQN improves the average reward by 4.01% over DQN.

Experimental Conclusion: IES improves policy diversity and adaptability, enabling the discovery of higher-reward routing strategies in dynamic environments.

3) Performance of DGA-IES Design: Methods training on Iridium topology and testing on Iridium-mod topology. Comparing different methods, including DQN method, DGA+DQN method, IES-DQN method, and DGA-IES method.

Results Analysis: As shown in Fig. 16(b), the proposed DGA and IES designs both enhance the agent's ability to find high-quality routing solutions under dynamic conditions. DGA improves state representation by addressing the over-smoothing problem of MPNN and capturing the complex non-Euclidean structure of satellite topologies. IES improves convergence speed and robustness by handling both periodic satellite movements and unpredictable link failures through parallel evolutionary training. When combined, the DGA-IES model achieves the best performance, with average reward improvements of 5.57% over DQN, 3.37% over DGA+DQN, and 1.52% over DQN+IES.

Experimental Conclusion: Both DGA and IES contribute independently to model effectiveness; their integration leads to further performance gains. The proposed DGA-IES method offers strong adaptability and efficient training, making it well-suited for complex and dynamic LEO satellite networks.

VII. Open Issues and Future Directions

Heterogeneous 6G Challenges for Space–Air–Ground Integrated Network: Strategies for effectively considering the impact of cross-domain heterogeneous resources (e.g., transmission, computation, and storage) on routing decisions.

Joint Learning Challenges of Multicategory Methods: Jointly exploiting the strengths of various learning methods to optimize distinct aspects of the routing process and achieve more robust solutions in complex interstellar environments.

Scalability and Synchronization Challenges in Multihop GNNs: Investigating the impact of multistep message aggregation on computation delay and synchronization overhead in real-time routing decisions. Explore asynchronous message-passing mechanisms and adaptive propagation depth control to balance decision quality with latency constraints in dynamic satellite networks.

VIII. Conclusion

This study proposes a novel DGA-IES approach, an intelligent routing method that integrates GNNs and RL to address dynamic routing challenges in LEO satellite networks. The DGA design effectively models complex network states within LEO constellations. In addition, by implementing parallel IES instead of conventional slow backpropagation training, the intelligent decision model converges rapidly in scenarios characterized by the high dynamics of interplanetary motion, adapting well to the dynamic nature of the LEO environment. To the best of our knowledge, this is the first application of ES for optimizing LEO satellite networks. Experimental results indicate that the proposed DGA-IES method can execute network routing tasks in dynamic satellite environments with lower E2E delays, reduced packet loss rates, and enhanced throughput performance. These improvements are crucial for ensuring seamless operation and maintaining QoS standards for resource-constrained satellite nodes.

Appendix A

Complexity Analysis of Algorithm 1: The time complexity of Algorithm 1 primarily arises from the message passing and message aggregation processes. In a single round of message passing, the range covers L-hop neighboring nodes. Assuming the worst case scenario where L-hop encompasses the entire topology, messages need to be passed along all edges $i \in \mathcal{E}$ within a single round. Thus, the time complexity of a single round of message passing in the worst case is $O(|\mathcal{E}|)$. The message passing process runs for T rounds, and during each round, every edge requires message passing. Since the 1×1 convolution operation can be parallelized across all nodes, the overall time complexity of the message passing process is $O(T|\mathcal{E}|)$.

The message aggregation process performs operations identical to the message passing process and shares the same order of magnitude. Therefore, its time complexity is also $O(T|\mathcal{E}|)$.

In conclusion, the overall time complexity of Algorithm 1 is $O(T|\mathcal{E}|)$.

APPENDIX B

Complexity Analysis of Algorithm 2: The time complexity of Algorithm 2 primarily stems from the parallel training process of agents.

For each agent, it only needs to complete the update of local parameters and upload the updated parameters to the coordinator. Thus, the time complexity for a single agent is $O(1)$. The coordinator is responsible for receiving the local parameters from all n agents and performing aggregation, resulting in a time complexity of $O(n)$ for the coordinator. Since the update processes of all agents are conducted in parallel, the overall time complexity of the algorithm 2 is $O(n)$.

REFERENCES

[1] C. Zhou et al., "Deep reinforcement learning for delay-oriented IoT task scheduling in SAGIN," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 911–925, Feb. 2021.

[2] G. Zeng, Y. Zhan, and H. Xie, "Channel allocation for mega LEO satellite constellations in the MEO-LEO networked telemetry system," *IEEE Internet Things J.*, vol. 10, no. 3, pp. 2545–2556, Feb. 2023.

[3] A. Ahmed, A. Al-Dweik, Y. Iraqi, and E. Damiani, "Integrated terrestrial-wired and LEO satellite with offline bidirectional cooperation for 6G IoT networks," *IEEE Internet Things J.*, vol. 8, no. 9, pp. 1767–1782, May 2024.

[4] Z. Rao, Y. Xu, Y. Yao, and W. Meng, "DAR-DRL: A dynamic adaptive routing method based on deep reinforcement learning," *Comput. Commun.*, vol. 228, Dec. 2024, Art. no. 107983.

[5] A. Shaikh and A. G. Greenberg, "OSPF monitoring: Architecture, design, and deployment experience," in *Proc. Symp. Netw. Syst. Des. Implement.*, 2004, pp. 57–70.

[6] I. D. Chakeres and E. M. Belding-Royer, "AODV routing protocol implementation design," in *Proc. Int. Conf. Distrib. Comput. Syst.*, Tokyo, Japan, 2004, pp. 698–703.

[7] L. Cui, F. R. Yu, and Q. Yan, "When big data meets software-defined networking: SDN for big data and big data for SDN," *IEEE Netw.*, vol. 30, no. 1, pp. 58–65, Jan./Feb. 2016.

[8] L. Zhou et al., "DPGNN: Dual-perception graph neural network for representation learning," *Knowl.-Based Syst.*, vol. 268, Aug. 2023, Art. no. 110377.

[9] Y. Cheng, J. Lu, D. Niyato, B. Lyu, J. Kang, and S. Zhu, "Federated transfer learning with client selection for intrusion detection in mobile edge computing," *IEEE Commun. Lett.*, vol. 26, no. 3, pp. 552–556, Mar. 2022.

[10] Y. Cheng, J. Lu, D. Niyato, B. Lyu, M. Xu, and S. Zhu, "Performance analysis and power allocation for covert mobile edge computing with RIS-aided NOMA," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4212–4227, May 2024.

[11] Z. Rao, Y. Xu, and S. Pan, "An intelligent routing method based on network partition," *Comput. Commun.*, vol. 160, pp. 25–33, Jul. 2020.

[12] X. Wang, Z. Dai, and Z. Xu, "LEO satellite network routing algorithm based on reinforcement learning," in *Proc. IEEE Int. Conf. Electron. Technol.*, Chengdu, China, 2021, pp. 1105–1109.

[13] S. Mahboob and L. Liu, "Revolutionizing future connectivity: A contemporary survey on AI-empowered satellite-based non-terrestrial networks in 6G," *IEEE Commun. Surveys Tuts.*, vol. 36, no. 2, pp. 1279–1321, 2nd Quart., 2024.

[14] B. Soret, I. Leyva-Mayorga, F. Lozano-Cuadra, and M. D. Thorsager, "Q-learning for distributed routing in LEO satellite constellations," in *Proc. Symp. Netw. Syst. Des. Implement.*, Stockholm, Sweden, 2024, pp. 63–70.

[15] Y. Lyu, H. Hu, R. Fan, Z. Liu, J. An, and S. Mao, "Dynamic routing for integrated satellite-terrestrial networks: A constrained multi-agent reinforcement learning approach," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1204–1218, May 2024.

[16] Z. Rao, Y. Xu, S. Pan, J. Guo, Y. Yan, and Z. Wang, "Cellular traffic prediction: A deep learning method considering dynamic nonlocal spatial correlation, self-attention, and correlation of spatiotemporal feature fusion," *IEEE Trans. Netw. Service Manag.*, vol. 20, no. 1, pp. 426–440, Mar. 2023.

[17] A. A. Magadum, A. Ranjan, and D. G. Narayan, "DeepQoSR: A deep reinforcement learning based QoS-aware routing for software defined data Center networks," in *Proc. Int. Conf. Comput. Commun. Netw. Technol.*, Kharagpur, India, 2021, pp. 1–7.

[18] J. Huang, Y. Yang, J. Lee, D. He, and Y. Li, "Deep reinforcement learning-based resource allocation for RSMA in LEO satellite-terrestrial networks," *IEEE Trans. Commun.*, vol. 72, no. 3, pp. 1341–1354, Mar. 2024.

[19] F. Dong, J. Song, Y. Zhang, Y. Wang, and T. Huang, "DRL-based load-balancing routing scheme for 6G space-air-ground integrated networks," *Remote. Sens.*, vol. 15, no. 11, pp. 2801–2822, Jul. 2023.

[20] W.-X. Liu, J. Cai, Q. C. Chen, and Y. Wang, "DRL-R: Deep reinforcement learning approach for intelligent routing in software-defined data-center networks," *J. Netw. Comput. Appl.*, vol. 177, Mar. 2021, Art. no. 102865.

[21] H. Liu, J. Lai, J. Zhu, L. Gan, and Z. Chang, "Enabling high-throughput routing for LEO satellite broadband networks: A flow-centric deep reinforcement learning approach," *IEEE Internet Things J.*, vol. 11, no. 17, pp. 28705–28720, Sep. 2024.

[22] L. Zhang, X. Yang, and Z. Bu, "Satellite network state prediction and route planning policy using the MPNN algorithm," in *Proc. ACM Int. Conf. Ser.*, Fukuoka, Japan, 2023, pp. 174–180.

[23] Q. He et al., "Routing optimization with deep reinforcement learning in knowledge defined networking," *IEEE Trans. Mobile Comput.*, vol. 23, no. 2, pp. 1444–1455, Feb. 2024.

[24] Y. Cheng, C. Ouyang, Y. Liu, and G. K. Karagiannidis, "On the performance of pinching-antenna systems (PASS) with orthogonal and non-orthogonal multiple access," 2025, *arXiv:2506.02420*.

[25] P. Xu, M. Feng, J. Zhou, L. Xiao, P. Xiao, and T. Jiang, "Inter-satellite routing for LEO satellite networks: A GNN and DRL integrated approach," in *Proc. IEEE/CIC Int. Conf. Commun.*, Chengdu, China, 2024, pp. 35–43.

[26] Y. Shi, W. Wang, X. Zhu, and H. Zhu, "Low earth orbit satellite network routing algorithm based on graph neural networks and deep Q-network," *Appl. Sci.*, vol. 14, no. 9, pp. 3840–3857, Apr. 2024.

[27] A. M. ud din and S. Qureshi, "Limits of depth: Over-smoothing and over-squashing in GNNs," *Big Data Min. Anal.*, vol. 7, no. 1, pp. 205–216, Jul. 2024.

[28] Z. Rao, Y. Xu, and S. Pan, "A deep learning-based constrained intelligent routing method," *Peer-to-Peer Netw. Appl.*, vol. 14, no. 4, pp. 2224–2235, Jun. 2021.

[29] J. W. Guck, A. V. Bemten, M. Reisslein, and W. Kellerer, "Unicast QoS routing algorithms for SDN: A comprehensive survey and performance evaluation," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 1, pp. 388–415, 1st Quart., 2018.

[30] Y. Gong, H. Yao, J. Wang, L. Jiang, and F. R. Yu, "Multi-agent driven resource allocation and interference management for deep edge networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 2018–2030, Feb. 2022.

[31] W. Jiang, "Software defined satellite networks: A survey," *Digit. Commun. Netw.*, vol. 9, no. 6, pp. 1243–1264, Jan. 2023.

[32] X. Cao, Y. Li, X. Xiong, and J. Wang, "Dynamic routings in satellite networks: An overview," *Sensors*, vol. 22, no. 12, p.4552, Sep. 2022.

[33] Z. Rao, Z. Zhu, Y. Yao, Y. Xu, Y. Cheng, and H. Du, "Computation-offloading optimization for satellite edge computing via diffusion and Lyapunov-based deep reinforcement learning," *IEEE Internet Things J.*, vol. 12, no. 18, pp. 37747–37762, Sep. 2025.

[34] J.-H. Lee, J. Park, M. Bennis, and Y.-C. Ko, "Integrating LEO satellites and multi-UAV reinforcement learning for hybrid FSO/RF non-terrestrial networks," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 3647–3662, Mar. 2023.

[35] B. Mao, X. Zhou, J. Liu, and N. Kato, "On an intelligent hierarchical routing strategy for ultra-dense free space optical low earth orbit satellite networks," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1219–1230, May 2024.

[36] G. R. Redinbo, "Queueing systems, volume I: Theory-LEOnard Kleinrock," *IEEE Trans. Commun.*, vol. 25, no. 1, pp. 178–179, Jan. 1977.

[37] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, "Message passing neural networks," *Mach. Learn. Meets Quan. Phys.*, vol. 14, no. 1, pp. 199–214, Jun. 2020.

[38] B. Guo, H. Li, Z. Zhang, and Y. Yan, "Online network slicing for real time applications in large-scale satellite networks," in *Proc. IEEE Int. Conf. Commun*, 2023, pp. 5564–5569.

[39] C. Huang, X. Zhou, X. Ran, Y. Liu, W. Deng, and W. Deng, "Co-evolutionary competitive swarm optimizer with three-phase for large-scale complex optimization problem," *Inf. Sci.*, vol. 619, pp. 2–18, Jan. 2023.

[40] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," *Mach. Learn.*, vol. 8, pp. 229–256, May 1992.

[41] Y. Cheng, K. H. Li, Y. Liu, K. C. Teh, and G. K. Karagiannidis, "Non-orthogonal multiple access (NOMA) with multiple intelligent reflecting surfaces," *IEEE Trans. Wireless Commun.*, vol. 20, no. 11, pp. 7184–7195, Nov. 2021.

[42] C. Güemes-Palau et al., "Accelerating deep reinforcement learning for digital twin network optimization with evolutionary strategies," in *Proc. NOMS*, 2022, pp. 1–5.

[43] Z. Wang, C. Chen, and D. Dong, "Instance weighted incremental evolution strategies for reinforcement learning in dynamic environments," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 9742–9756, Dec. 2023.

[44] O. M. Poncea, S. A. Pistirica, F. Moldoveanu, and V. Asavei, "Design and implementation of an openflow SDN controller in NS-3 discrete-event network simulator," *Int. J. High Perform. Comput. Netw.*, vol. 14, no. 1, pp. 17–29, Dec. 2019.

[45] G. Brockman et al., "OpenAI gym," 2016, *arXiv:1606.01540*.

[46] C. R. Harris et al., "Array programming with NumPy," *Nature*, vol. 585, no. 7825, pp. 357–362, Sep. 2020.

[47] D. D, S. E, and Šerifović–Trbalić A, "Performance of some image processing algorithms in tensorflow," in *Proc. Signal. Imag. Process.*, Maribor, Slovenia, 2018, pp. 1–4.

[48] A. Hagberg and D. Conway, "NetworkX: Network analysis with Python," 2020 [Online]. Available: https://networkx.github.io

[49] L. Dalcín and Y. L. Fang, "mpi4py: Status update after 12 years of development," *Comput. Sci. Eng.*, vol. 23, no. 4, pp. 47–54, Jul.-Aug. 2021.

[50] Y. Zhu and D. Lu, "Latency guaranteed joint optimal traffic intelligent scheduling in large-scale satellite networks," *Comput. Netw.*, vol. 242, Apr. 2024, Art. no. 110236.

[51] G. Chen, S. Qi, F. Shen, Q. Zeng, and Y.-D. Zhang, "Information-aware driven dynamic LEO-RAN slicing algorithm joint with communication, computing, and caching," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1044–1062, May 2024.

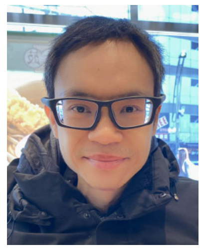
[52] J. Lee, Y. Cheng, D. Niyato, Y. L. Guan, and G. David González, "Intelligent resource allocation in joint radar-communication with graph neural networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 11120–11135, Oct. 2022.

Zheheng Rao (Member, IEEE) received the Ph.D. degree in communication and information system from the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University, Wuhan, China, in 2022.

He is currently a Lecturer with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou, China. His research interests include network intelligent management, cybersecurity, and machine learning.

Zhenyu Zhu received the B.E. degree in network engineering from Jiaxing Nanhu University, Jiaxing, China, in 2023. He is currently pursuing the M.E. degree with the College of Cyberspace Security, Hangzhou Dianzi University, Hangzhou, China.

His research interests mainly focus on machine learning and satellite communications.



Dusit Niyato (Fellow, IEEE) received the B.Eng. degree from the King Mongkuts Institute of Technology Ladkrabang, Bangkok, Thailand, in 1999, and the Ph.D. degree in electrical and computer engineering from the University of Manitoba, Winnipeg, MB, Canada, in 2008.

He is currently a Professor with the College of Computing and Data Science, Nanyang Technological University, Singapore. His research interests are in the areas of mobile generative AI, edge general intelligence, quantum computing and networking, and incentive mechanism design.

Ye Yao (Member, IEEE) received the M.S. degree in computer science and the Ph.D. degree in communication and information systems from Wuhan University, Wuhan, China, in 2005 and 2008, respectively.

He was a Visiting Scholar with the New Jersey Institute of Technology, Newark, NJ, USA, from December 2016 to December 2017. He is currently a Professor with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou, China. His research interests include multimedia forensics and information security.

Yanyan Xu received the Ph.D. degree in communication and information systems from Wuhan University, Wuhan, China, in 2007.

She is currently a Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing, Wuhan University. Her research interests include geospatial information security and intelligent network communication.

Yanyu Cheng (Senior Member, IEEE) received the B.S.Eng. degree in information engineering from Shanghai Jiao Tong University, Shanghai, China, in 2015, and the M.Sc. degree in signal processing and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University (NTU), Singapore, in 2016 and 2021, respectively.

From 2021 to 2023, he was a Research Fellow with the Alibaba-NTU Singapore Joint Research Institute, NTU. He is currently a Tenure-Track Professor with the School of Cyberspace, Hangzhou Dianzi University, Hangzhou, China. His research interests include wireless communications, cybersecurity, and machine learning.